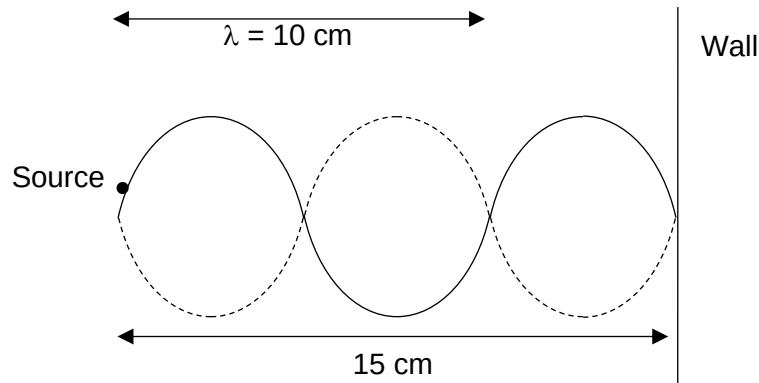


2(a) When the sound wave hits the wall, it will be reflected in the opposite direction and interfere with the original wave. [APPLICATION]

Now we have 2 waves of the same amplitude, frequency and wavelength moving in opposite directions interfering with each other [APPLICATION], giving rise to a stationary wave. [CONCLUSION]

[CONCEPT: conditions for formation of stationary wave]

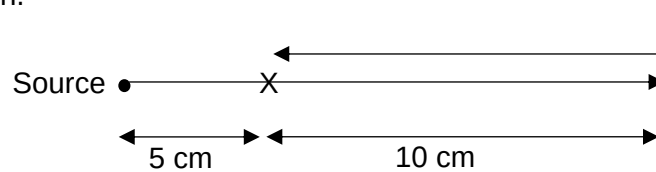
(b)



(c) From above diagram, $\lambda = 10 \text{ cm}$

$$\text{Frequency } f = \frac{v}{\lambda} = \frac{360}{0.10} = 3600 \text{ Hz}$$

d(i) After hitting the wall, the wave will continue to spread out as before, but in the opposite direction.



From diagram, reflected wave travels 25 cm.

$r_1 = 5 \text{ cm}$, amplitude $A_1 = 3.0 \times 10^{-5} \text{ m}$
 $r_2 = 25 \text{ cm}$, $A_2 = ?$

$$\text{Intensity } I \propto A^2 \propto \frac{1}{r^2} \text{ [CONCEPT]}$$

$$\therefore A \propto \frac{1}{r}$$

$$\frac{A_2}{A_1} = \frac{r_1}{r_2}$$

$$A_2 = A_1 \frac{r_1}{r_2}$$

$$= 3.0 \times 10^{-5} \times \frac{5}{25}$$

$$= 6.0 \times 10^{-6} \text{ m}$$

- (ii) Since X was originally a node, it is a place of destructive interference.
∴ resultant amplitude = $(3.0 - 0.6) = 2.4 \times 10^{-5} \text{ m}$

3(a) The electric field strength at a point is defined as the **electric force exerted per unit**